CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · ONCOLOGY PHASE 2,3 · STARTS 2015–2024

CTLA-4: 3 molecules stopped early, all monoclonal antibodies

11 stopped trials, but 3 molecules — all monoclonal antibodies: Ipilimumab, Erfonrilimab and Tremelimumab. Against 5.2% for all oncology trials with a resolved drug target and 4.5% across all oncology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

3.6%
11 of 307 closed trials (95% CI 2.0%–6.3%)

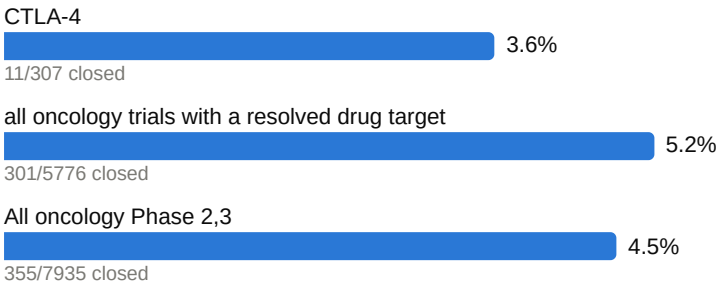
10 / 1 / 0
efficacy / safety / both (adds to 11)

40.4%
of 759 trials have closed · 452 still open or unresolved

1.1%
stopped for a biological reason within 3 years of starting (95% CI 0.5%–2.4%; 408 still at risk; 1.5% area-wide)

11 of 20 **Completed and missed the primary endpoint.** Of 232 completed trials in this segment, 20 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 11 missed and 7 met, 2 split across co-primary endpoints. Counted separately and never in the rate above; a floor, not a rate. 77 of the 210 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

How this compares



What is in this class: 285 distinct experimental drugs across 759 trials (antibody 753, small molecule 251, unknown 92, adc 32). Most tested: IPILIMUMAB (348), NIVOLUMAB (294), CADONILIMAB (138), DURVALUMAB (129), TREMELIMUMAB (129). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 11 stops came from 11 sponsor-asset programmes across 9 sponsors; the largest (Weill Medical College of Cornell University) contributed 1; removing that programme's trials from both sides leaves 3.3%. Screened alongside 85 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 9 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 6.5%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	0	142	0.0%	0.0%–2.6%
2018–2020	8	113	7.1%	3.6%–13.4%
2021–2024	3	52	5.8%	2.0%–15.6%

Where the stops sit

Efficacy / futility: 10, Safety: 1.
Sponsors with most stops: Herlev Hospital (2), Jiangsu Alphamab Biopharmaceuticals Co., Ltd (2), M.D. Anderson Cancer Center (1), Weill Medical College of Cornell University (1), University of Washington (1), H. Lee Moffitt Cancer Center and Research Institute (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT03274258	2	M.D. Anderson Cancer Center	IPILIMUMAB, NIVOLUMAB	2025-11	Efficacy	The study met its prespecified futility criteria during the prespecified interim futility analysis after enrolling the first cohort of 10 patients.

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT05116917	2	Herlev Hospital	NIVOLUMAB, IPILIMUMAB, Influenza vaccine	2023-10	Efficacy	Based on the precpecified interim analysis the predictive probability was below 10% for the prespecified endpoint.
NCT04925947	2	Weill Medical College of Cornell University	ERFONRILIMAB	2023-05	Efficacy	Data collected did not support the study endpoints
NCT05001724	3	Jiangsu Alphamab Biopharmaceuticals Co., Ltd	ERFONRILIMAB, LENVATINIB	2023-02	Efficacy	Because the efficacy data did not meet expectations, the sponsor decided to terminate the study.
NCT03838848	2	Jiangsu Alphamab Biopharmaceuticals Co., Ltd	ERFONRILIMAB, CARBOPLATIN, PEMETREXED, NINGETINIB	2022-07	Efficacy	Cohort A,B,C end enrollments. Cohort D and E were considered to have no significant clinical benefit at the SMC meeting and decided to terminate enrollment.
NCT03522584	2	University of Washington	DURVALUMAB, TREMELIMUMAB	2021-12	Efficacy	Terminated by the sponsor due to lack of efficacy in study treatment.

The 6 most recent of 11 stopped trials. Every molecule behind all 11 is named above; the complete trial list, the trials still running and the terminations with no readable cause ship with the dataset and the diligence report.

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 40.4% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta, in exchange for a work email, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.